

Setup system

Setup the system and get files from the internet.

In the following code, we will download the necessary files from the internet.

```

1 # SETUP: Please ignore this part.
2
3 # Remove existing files
4 import os
5
6 # Stop pygame from opening a window
7 os.environ["SDL_VIDEODRIVER"] = "dummy"
8
9 # Install kaleido package
10 !pip install -U kaleido==0.2.1
11
12 # Mount the drive:
13 from google.colab import drive
14 drive.mount('/content/drive')
15
16 try:
17     os.chdir("/content/drive/MyDrive/Race-2025")
18     print("Directory changed successfully.")
19 except FileNotFoundError:
20     print("Error: Race-2025 directory not found.")
21
22 # Create a local directory
23 !mkdir /content/drive/MyDrive/Race-2025
24
25 # Get the repo as a zip file and save it locally:
26 !wget https://github.com/pattichis/GraphSpeeds2/archive/refs/heads/main.zip -O repo.zip
27 !unzip repo.zip -d /content/drive/MyDrive/Race-2025/
28 !rm -f repo.zip
29
30
31 #!find './' -maxdepth 1 -type f -delete
32
33 # Use wget with -nc to avoid overwriting local copy.
34 !rm -f lineart_v4.py*
35 !wget https://raw.githubusercontent.com/pattichis/GraphSpeeds2/main/lineart_v4.py
36
37 # Download the pictures:
38 !wget -nc https://raw.githubusercontent.com/pattichis/GraphSpeeds2/main/Elephant.png
39 !wget -nc https://raw.githubusercontent.com/pattichis/GraphSpeeds2/main/Squirrel.png
40 !wget -nc https://raw.githubusercontent.com/pattichis/GraphSpeeds2/main/Tortoise.jpg
41 !wget -nc https://raw.githubusercontent.com/pattichis/GraphSpeeds2/main/Koala.jpeg
42
43 # Import the functions to use
44 from lineart_v4 import table, simulationVideo, textImage, plotTablesLines, CreateVideo, changeVideoSpeed

```

```

Requirement already satisfied: kaleido==0.2.1 in /usr/local/lib/python3.12/dist-packages (0.2.1)
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive", force_r
Directory changed successfully.
--2026-08-20 18:30:59-- https://raw.githubusercontent.com/pattichis/GraphSpeeds2/main/lineart_v4.py
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.109.133, 185.199.111.133, 185.199.110.13
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.109.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 34249 (33K) [text/plain]
Saving to: 'lineart_v4.py'

```

```

lineart_v4.py      100%[=====] 33.45K  --.-KB/s   in 0.005s

```

```

2026-08-20 18:30:59 (7.21 MB/s) - 'lineart_v4.py' saved [34249/34249]

```

File 'Elephant.png' already there; not retrieving.

```

--2026-08-20 18:30:59-- https://raw.githubusercontent.com/pattichis/GraphSpeeds2/main/Squirrel.png
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.111.133, 185.199.109.133, 185.199.110.13
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.111.133|:443... connected.
HTTP request sent, awaiting response... 404 Not Found
2026-08-20 18:31:00 ERROR 404: Not Found.

```

File 'Tortoise.jpg' already there; not retrieving.

File 'Koala.jpeg' already there; not retrieving.

Make videos I: Pick Find character images

Assignment: Find character images

1. Click on your Folder icon on the Left Hand Side.
2. Click on drive.
3. Click on MyDrive.
4. Click on Race-2025.
5. Click on GraphSpeeds2-main.
6. Look at the different images.
7. Write down the names of the image files that you want to use.

✓ Make videos II: Create tables for your characters

✓ Assignment: Create the tables and equations for your characters

1. Enter time and distance units in column_labels.
2. Enter time_values and distance_values for each value.
3. Enter names and equations for each character.

```

1 # Prepare the labels for everyone
2 column_labels = ["x (seconds)", "y (feet)"] # <- FIX units
3
4 # Table #1 values:
5 time_values = [0, 1, 2, 6.6, 13] # <- ENTER x values
6 distance_values = [0, 3.8, 7.7, 25, 50] # <- ENTER y values
7
8 # Create your table:
9 char1 = table(x=time_values, y=distance_values, column_labels=column_labels)
10 char1.plotTable()
11 char1.saveImage("char1.png")
12
13
14
15 # Table #2 values:
16 time_values = [0, 1, 2, 4.54, 9] # <- ENTER x values
17 distance_values = [0, 5.5, 11, 25, 50] # <- ENTER y values
18
19 # Create your table:
20 char2 = table(x=time_values, y=distance_values, column_labels=column_labels)
21 char2.plotTable()
22 char2.saveImage("char2.png")
23
24
25
26 # Table #3 values:
27 time_values = [0, 1, 2, 3.378, 7] # <- ENTER x values
28 distance_values = [0, 7.14, 14.8, 25, 50] # <- ENTER y values
29
30 # Create your table:
31 char3 = table(x=time_values, y=distance_values, column_labels=column_labels)
32 char3.plotTable()
33 char3.saveImage("char3.png")
34
35
36
37 # Table #4 values:
38 time_values = [0, 1, 2, 3, 4] # <- ENTER x values
39 distance_values = [0, 12.5, 25, 37.5, 50] # <- ENTER y values
40
41 # Create your table:

```

```
42 char4 = table(x=time_values, y=distance_values, column_labels=column_labels)
43 char4.plotTable()
44 char4.saveImage("char4.png")
45
46
47 # Merge all of the tables
48 tables_list = [char1, char2, char3, char4]
49
50 # Line equations of tables
51 legend_labels = [
52     'snoopy: y = 3.8*x', # <- Modify name and the equation here.
53     'Face: y = 5.5*x', # <- Modify name and the equation here.
54     'Minnie: y = 7.14*x', # <- Modify name and the equation here.
55     'Dog: y = 12.5*x'] # <- Modify name and the equation here.
56
57 # Plot tables
58 plotTablesLines(tables_list,
59                 fig_title = 'The Race',          # title on top
60                 x_label = 'Elapsed time (seconds)', # appears along x-axis
61                 y_label = 'Distance in feet',      # appears along y-axis
62                 legend_title = 'Journey',          # title on the right hand side
63                 legend_labels = legend_labels,     # text describing the table
64                 img_name = 'race.png')             # Saves under race.png.
```


x (seconds)	y (feet)
0.00000	0.00000
1.00000	3.80000
2.00000	7.70000
6.60000	25.00000
13.00000	50.00000

Wrote char1.png

x (seconds)	y (feet)
0.00000	0.00000
1.00000	5.50000
2.00000	11.00000
4.54000	25.00000
9.00000	50.00000

Wrote char2.png

x (seconds)	y (feet)
0.00000	0.00000
1.00000	7.14000
2.00000	14.80000
5.00000	25.00000
7.00000	50.00000

Make videos III: Specify names, images, and speeds

Assignment: place your characters

1. ENTER the names of the characters for your video.
2. ENTER filenames for each character in your video.

3. ENTER the speeds for your characters.

```

1 # Enter the names of your characters:
2 names = ["snoopy", "face", "Minnie", "Dog"]
3
4 # Enter the file names of your images:
5 images = ["snoopy.jpg", "face.jpg", "minnie.png", "dog.jpg"]
6
7 # Enter the speeds:
8 speeds = [3.8, 5.5, 7.14, 12.5]
9
10
11 # Directory:
12 directory = "/content/drive/MyDrive/Race-2025/GraphSpeeds2-main/"
13 images_list = [directory + filename for filename in images]
14
15 # Save all of the data:
16 data = [names, images_list, speeds]

```

3.00000	37.50000
4.00000	50.00000

Make videos IV: Make and experiment with your video

Your video will be stored to the video file `race.mp4` or a filename that you choose.

To make a video, one time unit (e.g., one hour) corresponds to one second of video.

If you set the `simulation_speed=2.0`, then one second will correspond to two time units (e.g., two hours).

For each second, the video contains a number of images. This is controlled by `fps` which means number of frames per second.

Assignment: experiment with your video

1. Enter the race distance, duration of the race, and video title.
2. Enter the units.
3. Change `simulation_speed` to 2.0. What do you observe?
4. Change `fps` to 20. What do you observe?
5. Save the video to a different file name.

The Race

```

1 # Experiment with the video
2 sim_video = simulationVideo(data = data,          # characters
3                             duration = 15,        # ENTER duration
4                             race_distance = 50,    # ENTER race distance
5                             vid_title = "Race")    # video title
6
7 sim_video.set_video(video_name="race.mp4",        # new filename
8                     fps=10,                        # frames per second
9                     simulation_speed=1.0)          # speed of the simulation
10
11 sim_video.set_units(distance_string="feet",        # ENTER distance unit.
12                     time_string="seconds",         # ENTER time unit.
13                     speed_string="feet/second")    # ENTER speed units.
14
15
16 # Create and save your video:
17 video = sim_video.create_video()
18 video.ipynthon_display()

```

```
File 'race.mp4' has been removed.
VideoWriter initialized successfully.
video file = race.mp4 closed.
Moviepy - Building video __temp__.mp4.
Moviepy - Writing video __temp__.mp4

Moviepy - Done !
Moviepy - video ready __temp__.mp4
```

0:00 / 0:14

✓ Make videos V: Make title and explanatory text images

You can create many text frames and save them into different frames.

This is done using `saveImage("filename.png")` as given below:

```
text = ""First line.
Second line.
Third line.""
textImage(frame1.png", text)
```

Assignment: Create text images

Fill out the explanatory text images.

```
1 # Create Title and group name here
2 text = ""Session 2 Group C:
3 ""
4 textImage("frame1.png", text)
5
6
7 # Who is racing?
8 text = ""Two dogs, a mouse,
9 and a head are racing!
10 ""
11 textImage("frame2.png", text)
12
13
14 # Why they are racing?
15 text = ""They are racing to see who is faster?
16 ""
```

```

17 textImage("frame3.png", text)
18
19
20 # snoopy equation and explanation:
21 text = """"Snoopy equation:
22 y = 3.8 * x
23 snoopy is going 3.8 feet in one sec
24 """"
25 textImage("frame4.png", text)
26
27
28 # Face equation and explanation:
29 text = """"Face equation:
30 y = 5.5 * x
31 Face is going 5.5 feet in one sec
32 """"
33 textImage("frame5.png", text)
34
35 # Minnie equation and explanation:
36 text = """"Minnie equation:
37 y = 7.14 * x
38 7.14 feet per sec
39 """"
40 textImage("frame6.png", text)
41
42
43 # Dog equation and explanation:
44 text = """"Dog's equation:
45 y = 12.5 * x
46 Dog is going 12.5 feet in one sec!
47 """"
48 textImage("frame7.png", text)
49
50
51 # Explain the winner:
52 text = """"Dog won because
53 he went 12.5 feet per second.
54 he can run so fast because thats
55 how he escapes reality.
56 50 / 4 = 12.5
57 """"
58 textImage("frame8.png", text)
59
60 # Victory page:
61 text = """"yay!
62 Dog won the race.
63 He is the fastest!
64 """"
65 textImage("frame9.png", text)

```

✓ Make videos VII: Make the final video

You can make the final video by combining images and videos together.

Assignment: Make final video

Verify that your video satisfies the requirements of your assignment!

```

1 # Confirm that you have all the image here:
2 img_list = ["frame1.png", "frame2.png", "frame3.png",
3             "race.mp4",
4             "char1.png", "frame4.png",
5             "char2.png", "frame5.png",
6             "char3.png", "frame6.png",
7             "char4.png", "frame7.png", "race.png",
8             "frame8.png", "frame9.png"]
9
10 # Create the video:
11 video=CreateVideo(video_name="all.mp4",      # This is file name of the final video.
12                   file_list=img_list,       # This the list of images and files to combine.
13                   fps=10,                   # The number of frames per second.
14                   duration=2.5)             # The duration for each image.
15
16 # Print the video name

```



```
16 # Display the video:
17 video.ipynon_display()
```

```
Opening frame1.png
Opening frame2.png
Opening frame3.png
Opening race.mp4
Opening char1.png
Opening frame4.png
Opening char2.png
Opening frame5.png
Opening char3.png
Opening frame6.png
Opening char4.png
Opening frame7.png
Opening race.png
Opening frame8.png
Opening frame9.png
video: height = 600 width = 800
File 'all.mp4' has been removed.
VideoWriter initialized successfully.
Moviepy - Building video __temp__.mp4.
Moviepy - Writing video __temp__.mp4

Moviepy - Done !
Moviepy - video ready __temp__.mp4
```

0:00 / 0:49

✓ SAVE YOUR VIDEOS AND GOOGLE COLAB NOTEBOOKS

Double-click (or enter) to edit

```
1 from moviepy.editor import concatenate_videoclips
2
3 clips = []
4
5 for i, filename in enumerate(img_list):
6
7     # Give EVERY clip its own unique temporary filename
8     temp_name = f"temp_{i}.mp4"
9
10    if filename.startswith("char"):
11        duration = 4.0
12
```